

FIFA 22 Player Market Value Analysis

Exploratory analysis, value estimation, Top-10% identification, and under-priced player screening

Executive Summary

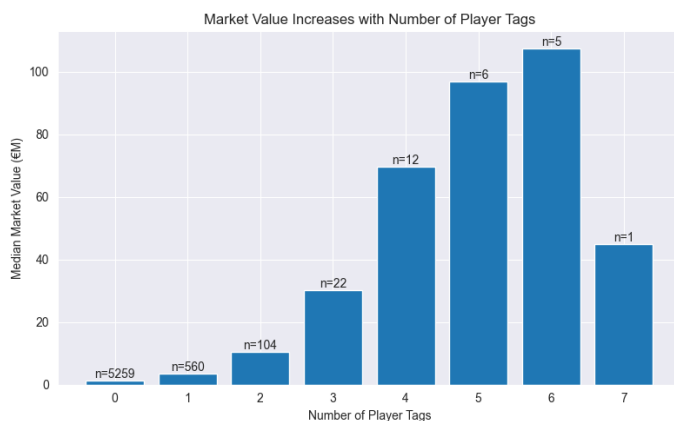
- Potential and Overall are the dominant indicators of market value. In the final model they account for roughly 46% and 42% of model importance, respectively.
- Age adds substantial information even among similarly rated players: younger players consistently command a premium, while technical attributes refine the estimate according to playing role.
- League context matters, but much less than player quality. Comparable Big Five players show a median value premium of about 16% after approximately controlling for age, Overall and Potential.
- The selected model identifies the held-out test set Top 10% with 99.2% precision and 99.2% recall (118 of 119 Top-10% players correctly identified).

1. Data Understanding, Cleaning and Preparation

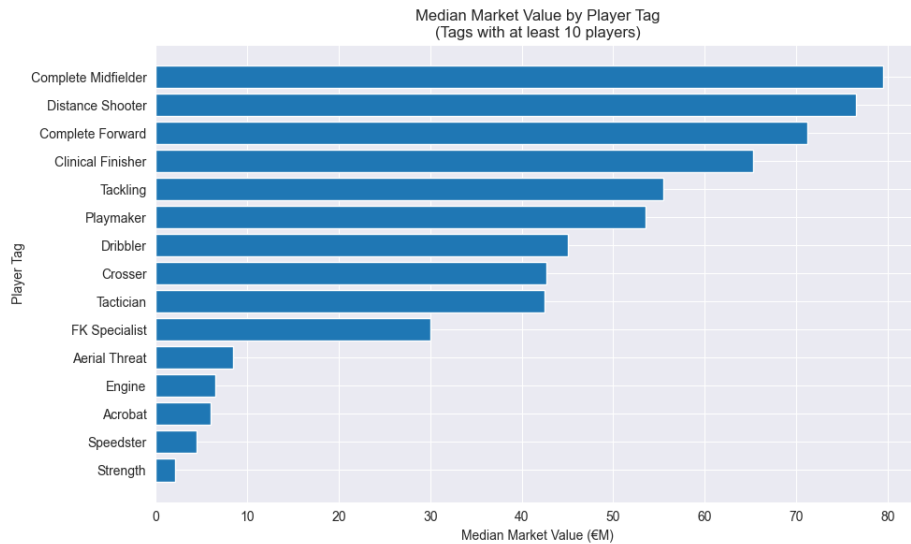
The provided FIFA 22 dataset contains 5,969 players and 21 original columns, with value_eur as the target variable. The analysis began with a data profile covering data types, missingness, duplicates, target distribution, extreme values and categorical-cardinality issues.

The following preparation steps were applied before modeling:

- **Removed** the redundant **index column** (Unnamed: 0) and converted **value_eur** from **text formatting to a numeric value**. The target contains no missing values.
- **Converted contract expiry year into contract_years_left**, using 2021 as the FIFA 22 reference year.
- Investigated missing values instead of applying blanket imputation. **The six outfield attributes** (pace, shooting, passing, dribbling, defending, physic) **are missing for exactly 543 players (9.1%), and all 543 are goalkeepers**. These are **structural missing values, not data errors**.
- **Parsed player_tags and created number_of_tags**. Players with **tags were substantially more valuable** on average than players without tags, **and some specific tags were associated with particularly high median market values**. However, the tag information was **highly sparse**: approximately **88% of players had no recorded tag**, while many individual tags appeared for only a small number of players. Encoding each tag separately would therefore create a large set of sparse categorical features and increase the risk of overfitting. Instead, the tag information was summarized using **number_of_tags**, which captures whether a player has accumulated multiple notable attributes while providing a more compact and generalizable feature. The relationship should be interpreted as an association rather than evidence that receiving a tag itself increases market value.



overall_band	has_tag	Players	Median Value (€M)
0 ≤64	0	1175	0.50
1 ≤64	1	27	0.50
2 65-69	0	1790	1.10
3 65-69	1	134	1.20
4 70-74	0	1412	2.40
5 70-74	1	180	2.50
6 75-79	0	690	8.50
7 75-79	1	183	10.50
8 80-84	0	170	25.00
9 80-84	1	120	30.75
10 85-89	0	19	45.00
11 85-89	1	58	68.25
12 90+	0	3	99.00
13 90+	1	8	122.50



- **Grouped leagues into Premier League, La Liga, Bundesliga, Serie A, Ligue 1 and Other** for a compact, more generalizable market-context feature.
- The profiling / duplicate checks found no duplicate rows and no duplicate sofifa_id values.

Data-quality item	Result
Players	5,969
Original columns	21
Missing target values	0
Duplicate rows / IDs	0 / 0
Goalkeepers with structural missing skills	543 (9.1%)
Players with no recorded tag	88.1%

Treatment of extreme market values

Market value is highly right-skewed. A standard IQR rule gives an upper bound of approximately €8.09M and flags 836 players (14.0%) as statistical outliers. These observations were retained: inspection showed that **the highest values belong to legitimate elite players** rather than obvious data errors. Removing them would discard the exact high-value segment the analysis is intended to understand.

Modeling split

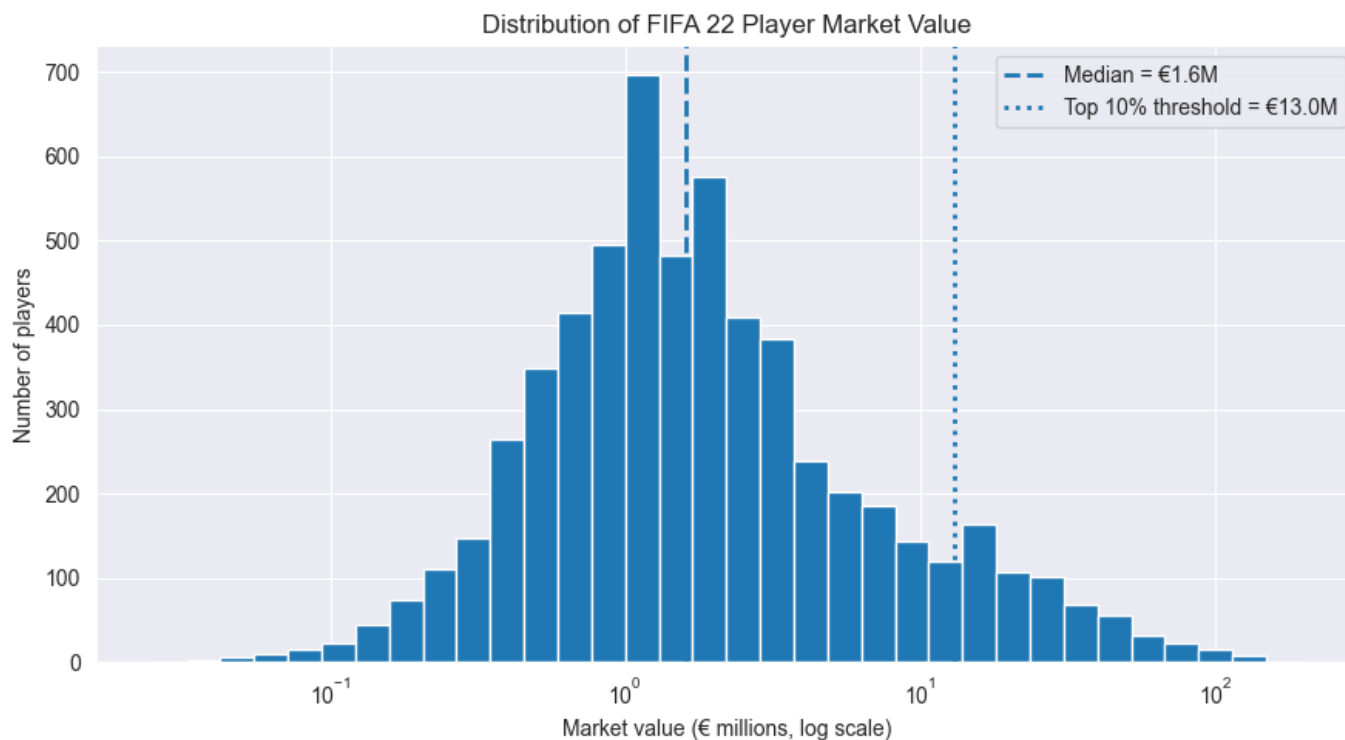
The data was split into **80% training and 20% held-out test data**. **Market-value quantile bins were used only to stratify the split, preserving a similar value distribution across both sets.**

Set	Players	Median value	Mean value
Training	4,775	€1.60M	€5.21M
Test	1,194	€1.60M	€5.15M

2. Exploratory Data Analysis: Three Key Takeaways

Takeaway 1 - Player value is highly concentrated in a small elite segment

The median player is worth about €1.6M, while the mean is about €5.2M, reflecting a strongly right-skewed market. In the full dataset, the empirical 90th percentile is €13M, and players at or above that level account for roughly 61% of total player value. This concentration motivates both the log transformation used later in modeling and the explicit focus on Top-10% identification.



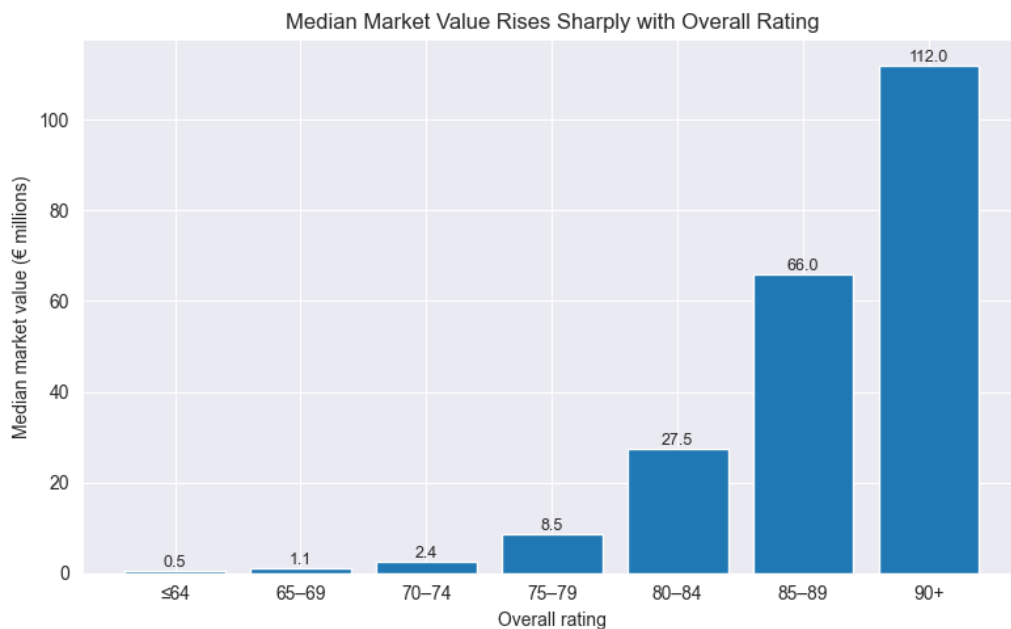
Distribution of market value. The log-scaled x-axis makes the long upper tail visible while retaining the median and Top-10% reference points.

Takeaway 2 - Player value increases nonlinearly with player quality

Overall and Potential are the strongest individual signals of player value, but their relationship with market value is not proportional. Market value rises increasingly sharply as players move into higher quality levels. For example, median value increases from approximately **€2.4M** for players rated 70–74 to around **€27.5M** for players rated 80–84.

This means that an improvement in player quality does not translate into a constant increase in market value across the rating range. The effect becomes much stronger among elite players, suggesting that value estimation should allow for nonlinear relationships and interactions between player characteristics.

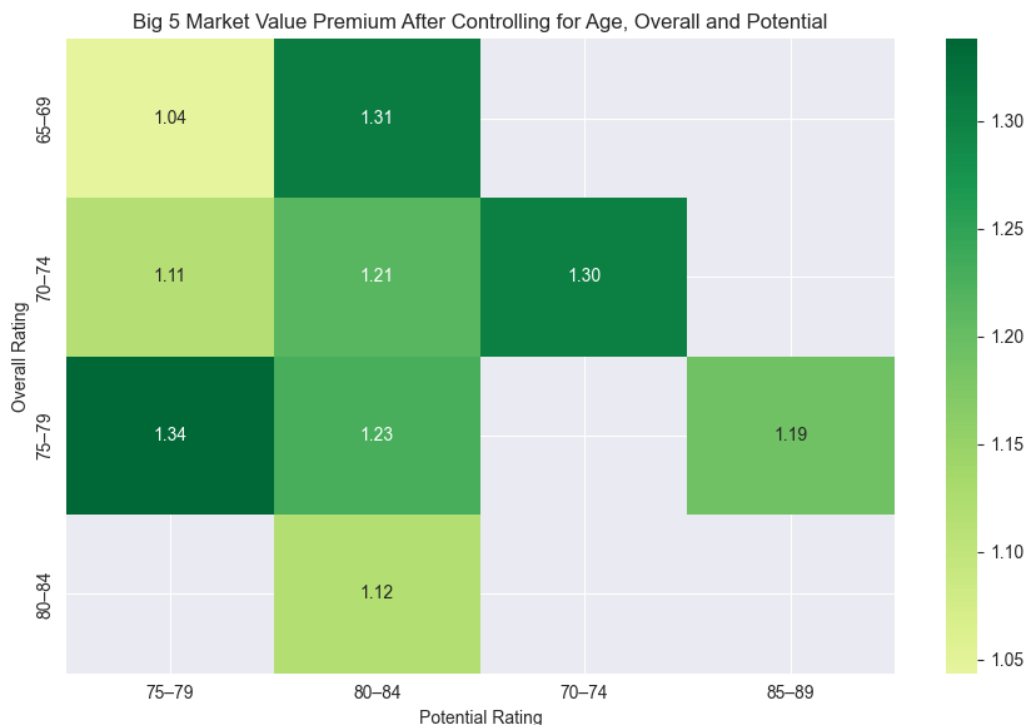
This EDA finding motivates the use of nonlinear models such as tree-based methods rather than relying only on a single linear relationship between player characteristics and market value.



Median market value across Overall rating groups. The increasingly steep rise in value at higher ratings illustrates that the relationship between player quality and market value is strongly nonlinear.

Takeaway 3 - League context adds value beyond basic player quality.

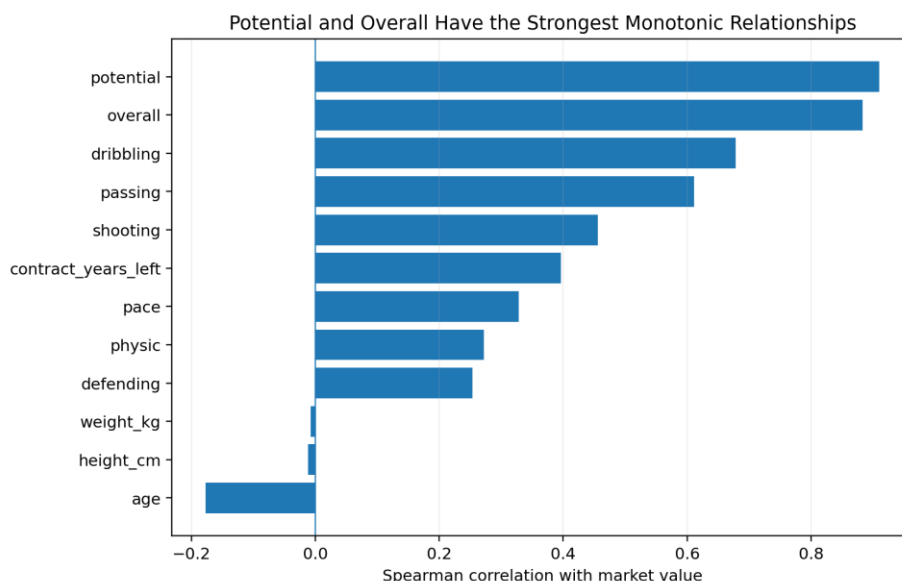
Even among players with broadly similar age, Overall and Potential, Big Five players generally command higher market values. Across the reliable comparison groups, the median Big Five premium was approximately 16%, although the size of the premium varies across player-quality levels.



Big Five market-value premium among comparable players. Each cell shows the median ratio between the market value of Big Five players and players from other leagues within comparable age, Overall and Potential groups. A value of 1.00 indicates no premium, while a value of 1.20, for example, indicates that Big Five players are valued approximately 20% higher within that comparison group. Age is controlled through age-matched comparisons and is therefore not shown as a separate axis.

3. Feature Selection for Value Estimation

Features were selected using the EDA, football interpretation, and expected ability to generalize to unseen players. The strongest numeric relationships are Potential (Spearman $\rho = 0.910$) and Overall ($\rho = 0.883$), while role-specific attributes become more informative within positions - e.g., shooting for attackers, defending for defenders, and passing/dribbling for midfielders.



Spearman rank correlations between numeric player attributes and market value. Height and weight show almost no monotonic relationship and were not retained.

Selected feature	Why it is relevant
overall	Compact measure of current player quality and one of the strongest observed value indicators.
potential	Captures future ceiling / upside and has the strongest monotonic relationship with market value.
age	Captures the strong youth premium observed even among players with similar current ability.
contract_years_left	Adds a limited but potentially useful contractual signal after the main player characteristics are accounted for.
pace	Role-relevant athletic/technical attribute, particularly useful for attacking and wide roles.
shooting	Adds attacking quality information and is especially informative for attackers.
passing	Captures creation and distribution ability; particularly informative for midfielders.
dribbling	Provides additional attacking and midfield quality information beyond Overall and Potential.
defending	Captures role-specific defensive ability and is particularly informative for defenders.
physic	Adds physical-profile information that can help distinguish players within similar roles and ratings.
formation	Represents broad playing role (att / mid / def / GK), which changes the interpretation of technical attributes.

Selected feature	Why it is relevant
league_group	Preserves market-context information while avoiding the sparsity and overfitting risk of the original many-category league variable.
number_of_tags	Compact signal for special player labels; tags are sparse and much of their raw value premium overlaps with player quality, so only the count is retained.

Excluded variables included identifiers and player names, club_name (high cardinality and risk of memorizing specific clubs), height and weight (near-zero correlations), and preferred foot (removed height, weight and preferred foot also because prior knowledge.). Goalkeeper structural missing values were handled naturally by the tree-based models; position-specific benchmark models used a reduced goalkeeper feature set.

4. Model Development and Selection

The modeling process was designed around two linked business needs: estimate a continuous market value for each player, and use that value estimate as a ranking score for identifying the most valuable 10%. A regression-first approach was chosen because the same model can support both Top-10% screening and the later under-pricing analysis. All model selection was performed on the training set; the held-out test set was opened only after the final model and hyperparameters were fixed.

Step 1 - Split the data and build a leakage-safe preprocessing pipeline

The 5,969 players were split into 4,775 training players (80%) and 1,194 held-out test players (20%). **Market-value quantile bins were used only to stratify** the split so that low, mid and high value players were represented similarly in both sets. Within model development, **five-fold cross-validation** was applied to the training set only. Categorical variables (formation and league_group) were **one-hot encoded** inside the pipeline, so each fold learned its preprocessing from the corresponding training portion rather than from validation data.

Result: the test set remained outside model fitting, hyperparameter tuning and model selection, while every validation score was produced from players not used to fit that fold.

Step 2 - Compare candidate tree models and target transformations

Random Forest and XGBoost were selected as nonlinear candidates because the EDA showed that market value rises sharply rather than proportionally with player quality. Each algorithm was evaluated twice: once on the original euro value and once on $\log(\text{value_eur})$. The log target was tested because the market-value distribution is strongly right-skewed and a few elite players sit far above the rest of the market. Hyperparameters were selected by five-fold cross-validated RMSE, while R^2 and MAE were reported as complementary measures.

Model	Target	R^2	MAE (€M)	RMSE (€M)
Random Forest	Raw	0.969	0.377	2.015
Random Forest	Log	0.978	0.337	1.703
XGBoost	Raw	0.985	0.317	1.387
XGBoost	Log	0.991	0.287	1.063

Result: XGBoost with the log-transformed target was the strongest tuned tree candidate ($R^2 = 0.991$, MAE = €0.287M, RMSE = €1.063M). The consistent improvement from Raw to Log for both algorithms supported keeping the log transformation for subsequent experiments.

Step 3 - Evaluate Top-10% identification during cross-validation

Because the final business task includes identifying the most valuable 10% of players, the candidate regression models were also evaluated on their ability to rank high-value players. Within each validation fold, the exact 10% highest actual values were compared with the exact 10% highest predicted values, and F1-score was calculated.

Configuration	Top-10% F1 (mean $\pm$ SD)
Random Forest - Raw	0.985 $\pm$ 0.011

Random Forest - Log	0.985 ± 0.005
XGBoost - Raw	0.985 ± 0.008
XGBoost - Log	0.983 ± 0.008

Result: all four tree configurations were effectively tied for Top-10% identification ($F1 \approx 0.983-0.985$). Ranking quality therefore did not provide a meaningful reason to prefer one of the best tree models, so price-estimation accuracy remained the main selection criterion. Under the exact Top-K definition, Precision and Recall are equal because the predicted and actual positive sets contain the same number of players.

Step 4 - Test whether position-specific specialization improves generalization

Motivation. The EDA showed that different technical attributes are informative for different playing roles - for example, shooting is more relevant for attackers, defending for defenders, and passing/dribbling for midfielders. This suggested that the relationship between player characteristics and market value may differ across positions, and that a single global model might be forcing one common pricing function across players with very different roles.

To test this, separate models were trained for attackers, midfielders, defenders and goalkeepers. Each player was evaluated only by the model corresponding to his formation. The predictions from the four position-specific models were then combined back into one unified prediction vector covering all training players, allowing the complete formation-specific system to be compared directly with the single global model.

Two specialization approaches were tested: formation-specific XGBoost Log models and formation-specific regularized linear models (Ridge, Lasso and Elastic Net) as an interpretable benchmark. For a fair comparison, all systems were evaluated using the same out-of-fold assignment, so the same players were held out in each comparison.

System	R ²	MAE (€M)	RMSE (€M)
Global XGBoost Log	0.988	0.295	1.304
Formation-Specific XGBoost Log	0.980	0.354	1.671
Formation-Specific Linear System	0.936	1.085	3.001

Result: the Global XGBoost Log system clearly remained best. Splitting by formation reduced the amount of training data available to each model and increased RMSE from €1.304M to €1.671M. The linear benchmark was substantially weaker (RMSE €3.001M), confirming that nonlinear relationships and interactions add important predictive information.

Why the RMSE differs from the tuning table: the €1.063M value above is the mean RMSE from the original candidate-model cross-validation used for tuning. The €1.304M value is a separate out-of-fold evaluation on the common fold assignment used for the apples-to-apples system comparison. They answer different development questions and are therefore not expected to be identical.

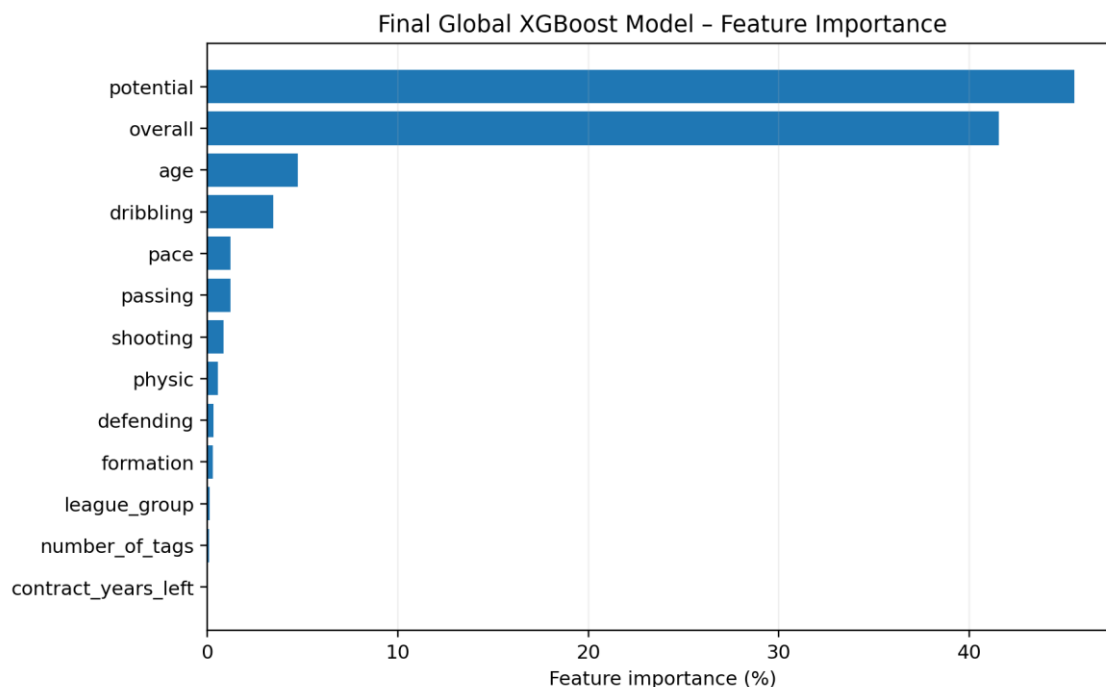
Step 5 - Lock the final model and refit it on the complete training set

The final selected model was Global XGBoost with a log-transformed target. The selected configuration used 400 trees, learning_rate = 0.03, max_depth = 4, min_child_weight = 3, subsample = 0.8, colsample_bytree = 1.0 and L2 regularization (reg_lambda = 1). After all development decisions were complete, this fixed model was trained once on all 4,775 training players.

Selected model: Global XGBoost with a log-transformed target. It delivered the best common out-of-fold regression performance and did not benefit from position-specific splitting.

What the final model uses most

After refitting, the encoded XGBoost feature importances were aggregated back to the original 13 input features. Potential and Overall dominate the model (roughly 46% and 42% of total importance), followed by Age and Dribbling. The remaining technical and context variables refine the estimate but contribute much less individually. These are model-internal importance scores and should not be interpreted as causal effects.



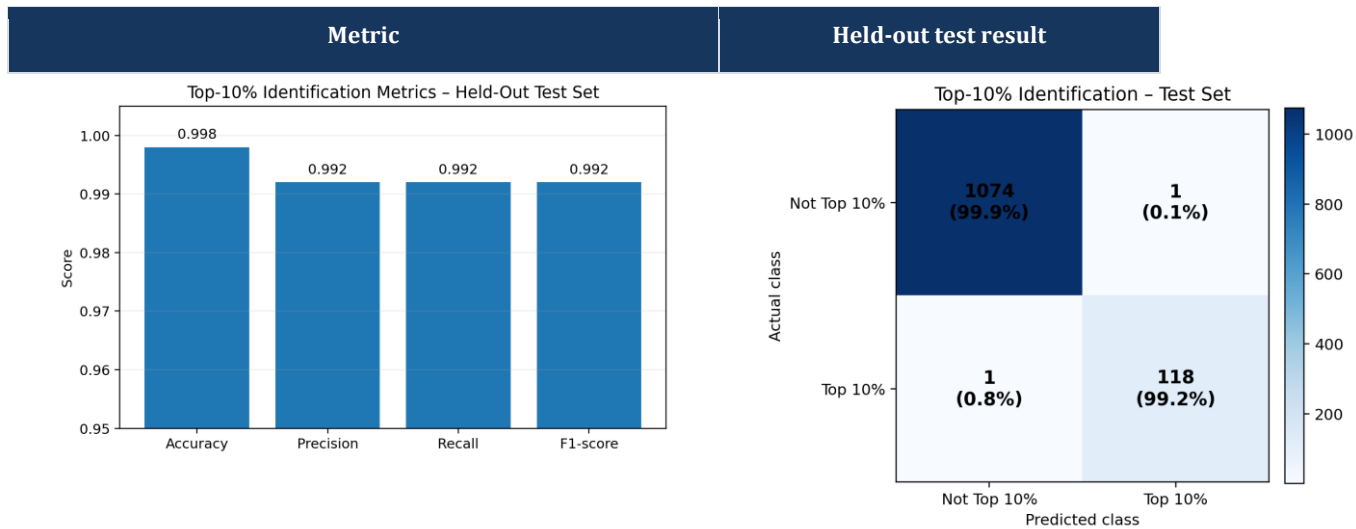
Aggregated feature importance from the final Global XGBoost Log model trained on the complete training set.

5. Top-10% Player Identification on the Held-Out Test Set

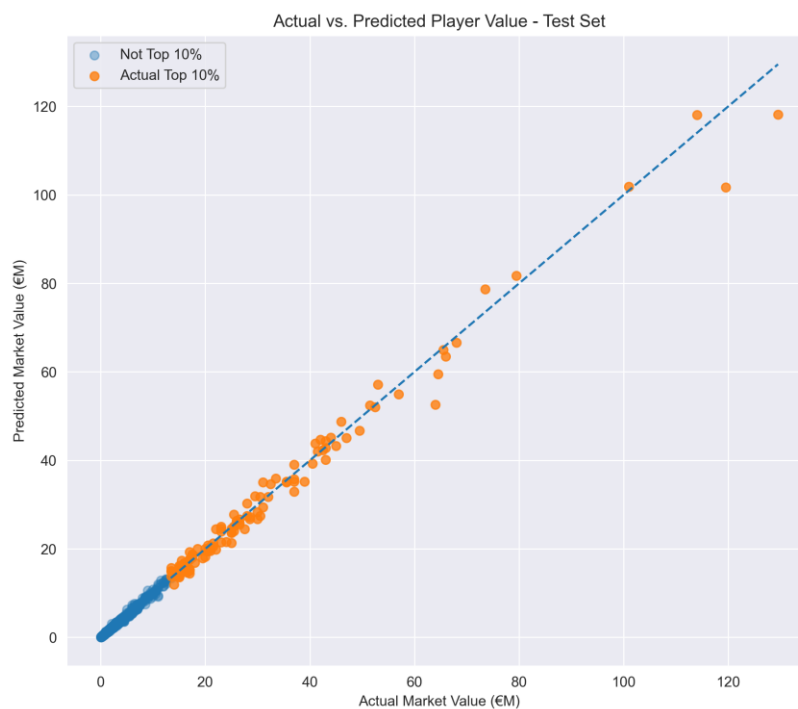
Once the final regression model was locked and refitted on all 4,775 training players, **it was applied to the 1,194 unseen test players**. The model predicted a market value for every test player, and these predicted values were used as ranking scores: **the 119 players with the highest predicted values were labeled as predicted Top-10%**, while **the exact 119 highest actual values formed the ground-truth Top-10%**. The classification is relative to the evaluated population.

With 1,194 test players, both actual and predicted Top-10% groups contain 119 players. Because the same number of positives is selected on each side, every false negative is paired with one false positive; therefore Precision and Recall are identical in this exact Top-K setup.

Metric	Held-out test result
Accuracy	0.998
Precision	0.992
Recall	0.992
F1-score	0.992
Average Precision	0.999



Held-out Top-10% evaluation. The confusion matrix contains 1,074 true negatives, 118 true positives, 1 false positive and 1 false negative.



Error analysis

Only two test players were on the wrong side of the exact Top-K cutoff: N. Milenković was a false negative (actual €14.0M, predicted €11.98M) and J. Vestergaard was a false positive (actual €13.5M, predicted €13.34M). The actual boundary contains a tie: five players are worth exactly €13.5M, while only four can enter the exact 119-player Top-10% set. This reinforces that the single swap occurs very near the boundary.

Interpretation: 118 of 119 actual Top-10% players were correctly identified. Average Precision of 0.999 indicates that the predicted value scores almost perfectly rank the true Top-10% players ahead of the rest of the test population.

6. Under-Priced Player Screening

Potential underpricing was evaluated only on the held-out test set, so every gap is based on an unseen-player prediction. Two rankings are useful: an absolute euro gap for the largest potential monetary opportunity, and a relative percentage gap for players who appear most undervalued compared with their observed price. These should be interpreted as model-based screening signals, not definitive evidence of market mispricing.

Top 10 by absolute underpricing gap

Player	Club	Pos.	Actual €M	Pred. €M	Gap €M
Fabinho	Liverpool	mid	73.50	78.63	5.13
J. Koundé	Sevilla FC	def	53.00	57.13	4.13
B. Kamara	Olympique de Marseille	mid	31.00	35.06	4.06
T. Alexander-Arnold	Liverpool	def	114.00	118.05	4.05
Palhinha	Sporting CP	mid	41.00	43.85	2.85
Raphinha	Leeds United	mid	46.00	48.76	2.76
Fabián	Napoli	mid	42.00	44.64	2.64
De Frutos	Levante Unión Deportiva	mid	22.00	24.50	2.50
V. Tsygankov	Dynamo Kyiv	mid	33.50	35.95	2.45
N. Mazraoui	Ajax	def	29.50	31.90	2.40

Top 10 by relative underpricing

Player	Club	Pos.	Actual €M	Pred. €M	Underpricing
Wagner Laranjo	Palmeiras	mid	2.10	3.07	46.25%
Vieirinha	PAOK	def	0.92	1.28	37.87%
J. Gol	Górník Łęczna	mid	0.16	0.22	34.61%
H. Kambou	Universidad Técnica de Cajamarca	def	0.07	0.09	32.08%
P. Skjelbred	Rosenborg BK	mid	1.00	1.30	30.37%
A. Johansson	Halmstads BK	def	0.18	0.23	30.17%
L. Montipò	Hellas Verona	GK	5.00	6.34	26.79%
F. Lemoine	FC Lorient	mid	1.00	1.26	25.67%
O. Alderete	Valencia CF	def	6.00	7.32	21.94%
A. Vlad	FCSB (Steaua)	GK	1.30	1.55	19.58%

7. Recommendations

- Use Potential, Overall and Age as the core valuation signals, then refine estimates using role-specific technical attributes and market context.
- Retain the Global XGBoost Log model for value estimation rather than splitting the market into position-specific models; the global model generalized better in out-of-fold testing.
- Use predicted market value as a ranking score for Top-10% screening. The held-out result (99.2% precision / recall) indicates that the approach is highly reliable for identifying elite-value players in this dataset.
- Treat underpricing lists as scouting leads rather than final decisions. They are useful for prioritization, but external information such as injuries, wage demands, transfer clauses and real-world market dynamics is not represented in the supplied dataset.

Bottom line: The analysis supports a compact business message: player quality and future upside drive most of the value signal; age materially changes what comparable ability is worth; and a tuned global XGBoost model can translate those relationships into both accurate value rankings and actionable player shortlists.